

ARXIV QUANTUM INFORMATION AND ALGORITHMS SURVEY

Section: Qubits and Quantum Superposition

A classical bit can exist in a definite state of either 0 or 1. A quantum bit or qubit can exist in a linear superposition of both orthogonal basis states: $|\psi\rangle = \alpha |0\rangle + \beta |1\rangle$, where α and β are complex probability amplitudes satisfying $|\alpha|^2 + |\beta|^2 = 1$.

Measurement collapses the superposition into a classical state $|0\rangle$ with probability $|\alpha|^2$ or $|1\rangle$ with probability $|\beta|^2$.

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Section: Quantum Entanglement and Bell States

Quantum entanglement is a phenomenon where quantum states of two or more particles cannot be described independently of each other. The canonical maximally entangled Bell state is $|\Phi^{++}\rangle = (|00\rangle + |11\rangle) / \sqrt{2}$.

Entanglement enables quantum teleportation, superdense coding, and exponential parallelism in quantum algorithms.